Neguse Solomon Mekonnen

Engineering • Technology • Research
About

Hello! 👋🏿

Thanks for stopping by! I'm Neguse Solomon, a civil and software engineer passionate about technology, research, and innovation. I enjoy combining engineering and software development to solve real-world problems and create practical solutions.

My Journey

My journey began with a BSc in Civil Engineering from Wollo University, Ethiopia, where I focused on structural design and sustainable construction.

I later moved to Europe for the Erasmus Mundus Joint Master's in Risk Assessment and Management of Civil Infrastructures (NoRISK), studying at the University of Padua in Italy and UPC Barcelona in Spain. My master's research focused on bridge structural reliability using Bayesian methods, monitoring data, and proof load testing.

Now, I'm pursuing a BSc in Computer Science at the University of the People, building my skills in software engineering and app development. I'm particularly interested in how engineering, software, and research can come together to solve real-world problems.

What I Bring to the Table

▪Structural Engineering and Infrastructure Design

▪Structural Reliability and Risk Assessment

▪Software Engineering and App Development

▪Computational Engineering and Scientific Programming

▪Engineering Research and Innovation

How I Work

▪I approach complex problems with analytical thinking and practical solutions.

▪I combine engineering knowledge, research, and software development to solve real-world challenges.

▪I value continuous learning, innovation, and interdisciplinary collaboration.

▪I focus on developing reliable, efficient, and sustainable solutions.

Let's Build the Future

I'm always interested in connecting with people who are curious about engineering, technology and research. I enjoy exchanging ideas, learning from others, and exploring opportunities to work together.

If any of these areas interest you too, let's connect. I'm always happy to have a good conversation and see where it leads.

Featured

Media

NoRisk, No Limits — M.Sc., Class of 2026

Completed the Erasmus Mundus Joint Masters in Risk Assessment and Management of Civil Infrastructures (NoRisk) from UNIPD and UPC marks more than an academic milestone; it refines my blueprint for impact.

This journey reflects a shift from building structures to safeguarding them against uncertainty. Risk management isn't just technical it’s a commitment to resilience, bridging European innovation to Africa’s infrastructural future.

Humbled by the rigor. Grateful for the borders crossed. Ready for the responsibility. 👷🌍

Edit

Delete

Media

The Face of the NORISK Master Programme

Months after joining the NORISK Master, the consortium chose my face for their global recruitment campaigns — urging applicants to apply and announcing admissions. From Ethiopia to Europe, I’m proud to represent African talent on this stage.

Edit

Delete

Media

My Impact Journey at EYEA — From Intern to Board Secretariat

A snapshot of my contributions at the Ethiopian Youth Entrepreneurs Association (EYEA), where I grew from Project Intern to Secretariat of the Board. Highlights include growing our

LinkedIn presence from 2K to 16K+, launching The Catalyst Magazine, leading communications for 10+ national events, and shaping EYEA's brand as a leading youth-led CSO in Ethiopia.

Edit

Delete

Media

Delegate at World Youth Development Forum 2025 in China CN

I had the honor of representing Ethiopia as the only Ethiopian delegate at the World Youth Development Forum 2025, held in Suzhou and Shanghai, China, from July 13–21. Alongside delegates from over 100 countries, I engaged in meaningful discussions on global development, youth empowerment, and international cooperation.

I was humbled to be awarded First Place in Cluster 10 – Partnership for the Goals, a moment that made this experience even more memorable. It strengthened my commitment to driving positive change and deepened my appreciation for global youth collaboration and shared responsibility for a better future.

Edit

Delete

Media

Launched The Catalyst Magazine

In Feb 2025, I launched The Catalyst — a magazine for Ethiopian youth entrepreneurs. Built to spotlight struggle and success, it’s already sparking conversations, shaping policy, and connecting young innovators across Africa.

Edit

Delete

Media

Delegate at Pan-African Youth Forum 2024 in Algeria DZ

I had the incredible honor of participating in the Pan-African Youth Forum 2024 in Oran, Algeria, from November 1st to 4th. Representing Ethiopia, I connected with passionate youth leaders from across Africa, discussing and collaborating on impactful solutions for the continent's future. Through this forum, I contributed to discussions on Youth Empowerment, innovative ideas, and sustainable growth.

My experience in Oran not only deepened my understanding of our shared African heritage but also strengthened my commitment to driving positive change in Africa. I explored Algeria’s rich history, built meaningful connections, and left with powerful memories that will guide my continued journey in youth leadership and pan-African collaboration.

Edit

Delete

Link

Featured in FBC (Fana Broadcasting Corporate S.C.) News

https://www.facebook.com/fanabroadcasting/posts/%E1%8B%88%E1%88%AE-%E1%88%99%E1%8D%88%E1%88%AA%E1%88%83%E1%89%B5-%E1%8A%AB%E1%88%9A%E1%88%8D-%E1%8A%A8%E1%8A%A2%E1%89%B5%E1%8B%AE%E1%8C%B5%E1%8B%AB-%E1%88%A5%E1%88%AB-%E1%8D%88%E1%8C%A3%E1%88%AA-%E1%8B%88%E1%8C%A3%E1%89%B6%E1%89%BD-%E1%8C%8B%E1%88%AD-%E1%89%B0%E1%8B%88%E1%8B%AB%E1%8B%A9-%E1%8A%A0%E1%8B%B2%E1%88%B5-%E1%8A%A0%E1%89%A0%E1%89%A3-%E1%88%98%E1%88%B5%E1%8A%A8%E1%88%A8%E1%88%9D-22-2017-%E1%8A%A4%E1%8D%8D-%E1%89%A2-%E1%88%B2-%E1%8B%A8%E1%88%A5%E1%88%AB%E1%8A%93-%E1%8A%AD%E1%88%85%E1%88%8E%E1%89%B5-%E1%88%9A/845133361152411/

I am honored to be featured in FBC news, highlighting a significant discussion with H.E. Muferihat Kamil, Minister of Labor and Skills, on the crucial role of youth entrepreneurship and skill development in Ethiopia. As a part of the Ethiopian Youth Entrepreneurs Association (EYEA), I am dedicated to advancing entrepreneurship and creating pathways for young leaders to contribute to Africa's sustainable growth. This moment reinforces our commitment to fostering innovation and empowering the next generation of entrepreneurs.

Edit

Delete

Link

Featured in H.E. Muferiat Kamil Facebook Page

https://www.facebook.com/muferihat.kamil.ahmed/posts/%E1%8B%A8%E1%88%A5%E1%88%AB-%E1%8D%88%E1%8C%A0%E1%88%AB-%E1%88%B5%E1%8A%90-%E1%88%9D%E1%88%85%E1%8B%B3%E1%88%A9%E1%8A%95-%E1%88%88%E1%8B%88%E1%8C%A3%E1%89%B6%E1%89%BD-%E1%8B%AD%E1%89%A0%E1%88%8D%E1%8C%A5-%E1%88%9D%E1%89%B9-

%E1%88%88%E1%88%9B%E1%8B%B5%E1%88%A8%E1%8C%8D-
%E1%8A%A2%E1%89%B5%E1%8B%AE%E1%8C%B5%E1%8B%AB-
%E1%8A%A5%E1%88%9D%E1%89%85-
%E1%8B%A8%E1%88%8D%E1%88%9B%E1%89%B5-
%E1%8D%80%E1%8C%8B%E1%8B%8E%E1%89%BD-
%E1%8B%AB%E1%88%8F%E1%89%B5-%E1%88%80%E1%8C%88%E1%88%AD-
%E1%8A%93%E1%89%B5-%E1%8B%AD%E1%88%85%E1%8A%95%E1%8A%95-
%E1%8A%A0%E1%89%85%E1%88%9D-
%E1%8A%A0%E1%88%8D%E1%88%9D/1063055048784647/

I am honored to be part of a successful discussion with H.E. Muferiat Kamil, where we focused on the importance of youth entrepreneurship and skill development in Ethiopia. As a part of the Ethiopian Youth Entrepreneurs Association (EYEA), I am proud to contribute to the growth of entrepreneurship in Africa, creating opportunities for young leaders to drive development.

Edit

Delete

Media

Built to Last — Officially a Civil Engineer, Class of 2023

Graduating with Distinction in Civil Engineering from Wollo University — Class of 2023 — marks more than just the end of a chapter; it’s the beginning of everything I aspire to build.

This milestone reflects years of dedication, teamwork, and a passion for sustainable development.

Civil Engineering is not just a profession—it’s a commitment to designing solutions that uplift communities and shape Africa’s future.

Experience

Tebib Labs logo

Founding Engineer

Tebib Labs

Oct 2026 - Present · 2 mos

Addis Ababa, Ethiopia

I lead software engineering and applied research, developing innovative solutions to solve real-world problems..

Present

Universitat Politècnica de Catalunya logo

Graduate Research Assistant

Universitat Politècnica de Catalunya

Mar 2026 - Sep 2026 · 7 mos

Barcelona, Catalonia, Spain

I conducted structural reliability research applying Bayesian methods to bridge monitoring data and proof load testing.

Ethiopian Youth Entrepreneurs Association (EYEA) logo

Ethiopian Youth Entrepreneurs Association (EYEA)

1 yr 8 mos

Addis Ababa, Ethiopia

Comms Lead & Jr Technical Advisor

Feb 2024 - Aug 2025 · 1 yr 7 mos

LinkedIn helped me get this job helped me get this job

I led EYEA’s communications, managed the comms team and magazine editorial work, while providing technical advice to youth entrepreneurs.

Project Intern

Jan 2024 - Feb 2024 · 2 mos

LinkedIn helped me get this job helped me get this job

I supported the Program Manager by coordinating with partners and youth entrepreneurs, organizing activities, and assisting with the Zelela 9 program.

Ministry of Innovation And Technology- Ethiopia logo

IGF National Advisory Group Rep

Ministry of Innovation And Technology- Ethiopia

Mar 2024 - Dec 2024 · 10 mos

Addis Ababa, Ethiopia

I provided strategic advice and representation, collaborating with stakeholders to shape inclusive digital policies under the MInT.

Ketema Mekonnen General Contractor logo

Site Engineer

Ketema Mekonnen General Contractor

Oct 2023 - Jan 2024 · 4 mos

Addis Ababa, Ethiopia

I oversee construction projects, ensuring safety, quality, and efficient execution through strong teamwork and civil engineering expertise.

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU) logo

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

3 yrs 11 mos

Kombolcha, Amhara Region, Ethiopia

President

Jul 2021 - Sep 2023 · 2 yrs 3 mos

I managed the day-to-day operations of the association, execute legal documents, drive initiatives, and provide direction for the executive team.

Vice President For External Affairs

Feb 2021 - Jul 2021 · 6 mos

I lead brand building, communications, and partnership growth, driving initiatives to strengthen stakeholder relationships.

Vice President For Academic Affairs

Nov 2019 - Feb 2021 · 1 yr 4 mos

I had primary leadership responsibilities for planning, implementing, and coordinating the educational schedule of the association.

Bamacon Real-Estate logo

Internship Trainee

Bamacon Real-Estate

Nov 2022 - Jan 2023 · 3 mos

Addis Ababa, Ethiopia

I managed BOQs, bar schedules, and quantity takeoffs while coordinating with site teams and supervising operations.

Education

Universitat Politècnica de Catalunya logo

Universitat Politècnica de Catalunya

Master's Degree, Risk Assessment and Management of Civil Infrastructures

Mar 2026 – Sep 2026

Dissertation

Università degli Studi di Padova logo

Università degli Studi di Padova

Master's Degree, Risk Assessment and Management of Civil Infrastructures

Oct 2025 – Mar 2026

Coursework

University of the People logo

University of the People

Bachelor's degree, Computer Science

Sep 2024 – Jan 2027

Grade: Cum Laude

Wollo University logo

Wollo University

Bachelor of Science, Civil Engineering

Oct 2018 – Sep 2023

Grade: Distinction

Licenses & certifications

Università degli Studi di Padova logo

Laurea Magistrale in Ingegneria Civile

Università degli Studi di Padova

Issued Sep 2026

Credential ID nt_EmFJXQ9qO70ddjZqjVQ

Show credential

Arizona State University logo

Data Analytics and Digital Transformation

Arizona State University

Issued Feb 2025

Hanic Interior Design and Software Tutor Institute logo
Interior Design Certification

Hanic Interior Design and Software Tutor Institute

Issued Jul 2023

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU) logo
SAP and SAFE Software Training

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Issued Jul 2023

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU) logo
Civil 3D and InRoads Software Training

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Issued May 2022

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU) logo

AutoCAD and ETABS Software Training

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Issued Jun 2021

DevTown logo

Hotstar Clone using HTML, CSS and JavaScript

DevTown

Issued Aug 2023

Show credential

Alison logo

Diploma in Project Management

Alison

Issued Jun 2023

Credential ID 1266-22777052

Show credential

CENGG Software Solution logo

Advanced AutoCAD Software Training

CENGG Software Solution

Issued Aug 2019

CENGG Software Solution logo

Quantity Surveying - Excel with AutoCAD Drawing

CENGG Software Solution

Issued Aug 2021

CENGG Software Solution logo

ETABS - Structural Analysis and Design

CENGG Software Solution

Issued Aug 2021

Udemy logo

Contracts Management in Construction Projects

Udemy

Issued Dec 2021

Credential ID UC-6873f172-f375-43df-a200-66e827524979

Show credential

Projects

Projects from connected apps sync automatically and can’t be edited on LinkedIn.

Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct

Mar 2026 – Sep 2026

During this research period, I developed a Bayesian MCMC framework to quantitatively update the structural reliability of deteriorating concrete bridges. By integrating instrumented proof load testing with mechanics-based stop criteria (CSCT/CSDT), I validated the safety of the ASR-affected Zijlweg Viaduct. The analysis demonstrated a significant reliability gain ($\beta = 5.33$), proving the structure exceeds modern safety targets and supporting sustainable asset management by avoiding unnecessary demolition.… more

Associated with Universitat Politècnica de Catalunya

Multi-Hazard Methodology for Building Stock Management

Oct 2025 – Mar 2026

As part of the EMJM NORISK NR6 project, our team developed a GIS-based framework to rank 248 buildings against four hazards. We created normalized risk indices, validated results against Italian NTC standards, and conducted a combined retrofit analysis.… more

Contributors: Pedro and Morris

Associated with Università degli Studi di Padova

Design and Structural Analysis of a Mixed-Use Building Using Sustainable and High-Performance Construction Materials

May 2023 – Sep 2023

This research focuses on the structural design and performance analysis of a mixed-use building using advanced construction materials and methodologies. The study integrates innovative techniques to enhance sustainability, cost efficiency, and structural resilience, contributing to modern engineering solutions.… more

Associated with Wollo University

Pavement Distress Analysis and Mitigation Strategies – Kombolcha City

May 2023 – Aug 2023

As part of the Highway Engineering III (CEng5241) course, I studied pavement distress in Kombolcha City. The work involved identifying distress types, analyzing causes, and proposing mitigation strategies. The findings provide insights for road maintenance and improvements.… more

Associated with Wollo University

Mizan-Tepi to Bonga Road Upgrading Design Project

Feb 2023 – May 2023

As part of the Integrated Design II course (CEng5191), this project involved designing the upgrading of the Mizan-Tepi to Bonga road. The study included surveying, geometric and pavement design, drainage system planning, and cost estimation to improve road infrastructure and connectivity.… more

Associated with Wollo University

Structural Design and Analysis of G+4 Building

May 2022 – Jul 2022

As part of the Integrated Design course (CEng4191), I analyzed and designed a G+4 building. The work included load calculations, seismic analysis, and reinforcement detailing using software like ETABS. The project ensured structural stability, safety, and compliance with standards.… more

Associated with Wollo University

Volunteering

UoPeople Ambassador

University of the People

Mar 2025 - Present · 1 yr 9 mos

Education

I promote UoPeople’s mission by spreading awareness, engaging in social media activities, and helping others access tuition-free higher education.

Volunteer, Special Monitoring Mission (SMM), Ethiopia

International Human Rights Commission-IHRC

Oct 2024 - Present · 2 yrs 2 mos

Civil Rights and Social Action

I monitor and report human rights issues in Ethiopia by collecting data, conducting field assessments, and supporting advocacy for justice and equality.

IEP Ambassador

Institute for Economics & Peace

Oct 2024 - Present · 2 yrs 2 mos

Human Rights

Completed a peacebuilding program, delivered a community talk, and continue supporting global peace in a 5,000+ ambassador network.

Teacher

Addis Ababa Education Bureau

Jun 2019 - Sep 2019 · 4 mos

Education

I teach academic and life lessons, support local teachers with activity-based methods, and manage lesson planning, tasks, grading, and training.

Volunteer

Wollo University Kombolcha Institute of Technology Charity Club

Oct 2018 - Sep 2023 · 5 yrs

Social Services

I helped raise funds for those in need, supported various causes, planted trees, participated in harvesting, and tutored elementary and high school students.

Instructor

Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Oct 2021 - Aug 2023 · 1 yr 11 mos

Education

I delivered hands-on training in SAP2000, SAFE, Advanced Excel, AutoCAD, and ETABS for civil engineering students.

Publications (2)

Google Scholar

Show publication

Research Gate

Honors & awards

EMJM Consortium Scholarship

Issued by University of Minho, University of Padua, University of La Rochelle, and Polytechnic University of Catalonia · Oct 2025

Associated with Università degli Studi di Padova

Awarded an EMJM Consortium Scholarship covering €10,000 in full-year tuition fees and health insurance.

UoPeople Scholarship

Issued by University of the People · Sep 2024

Associated with University of the People

Awarded a full scholarship worth $6,900, covering all assessment fees throughout my degree program.

Certificate of Appreciation for Academic Contribution

Issued by Ethiopian Civil Engineering Students Association Wollo Local Unit (ECESA-WOU) · Sep 2023

Associated with Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Certified for leading tutorial classes and mentoring students on mini-projects in civil engineering (2019–2023).

Leadership Excellence Award

Issued by Wollo University · Sep 2023

Associated with Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Recognized for four years of leadership (2019–2023) as VP and President, advancing student engagement, academics, and external relations.

Organizations

Ethiopian Youth Entrepreneurs Association (EYEA)

Secretariat of the Board · Jun 2025 – Present

Associated with Ethiopian Youth Entrepreneurs Association (EYEA)

I manage board documentation, organize meetings, prepare minutes, and coordinate communication to support effective governance.

Ethiopian Civil Engineering Students Association Wollo Local Unit (ECESA-WOU)

Honorary Member · Sep 2023 – Present

Associated with Ethiopian Civil Engineering Students Association – Wollo Local Unit (ECESA-WOU)

Former President and Vice President at ECESA-WOU; now support the association’s mission and student development as an honorary member.

American Society of Civil Engineers

Member · Jan 2023 – Present

Associated with Wollo University

Actively engage with global professionals, access valuable industry resources, and contribute to innovative engineering discussions and solutions.